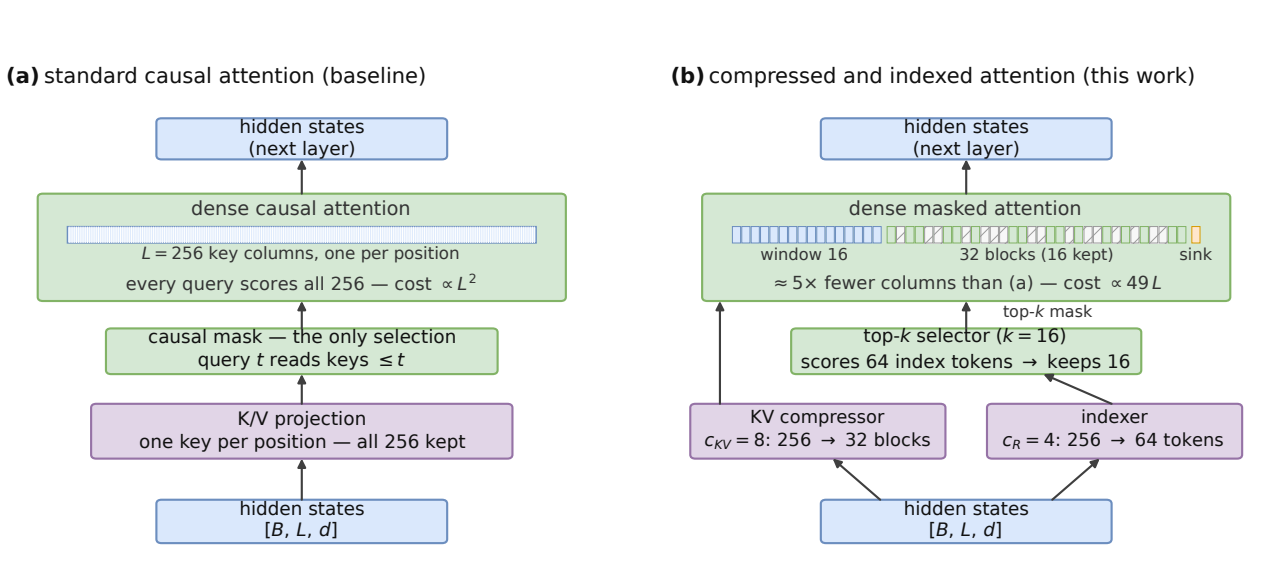
(a) standard causal attention (baseline)
hidden states
(next layer)
dense causal attention
L = 256 key columns, one per position
every query scores all 256 — cost ∝ L²
causal mask — the only selection
query t reads keys ≤ t
K/V projection
one key per position — all 256 kept
hidden states
[B, L, d]
(b) compressed and indexed attention (this work)
hidden states
(next layer)
dense masked attention
window 16
32 blocks (16 kept)
sink
≈ 5× fewer columns than (a) — cost ∝ 49L
top-k mask
top-k selector (k = 16)
scores 64 index tokens → keeps 16
KV compressor
c_KV = 8: 256 → 32 blocks
indexer
c_R = 4: 256 → 64 tokens
hidden states
[B, L, d]